

Nanoelectronics - Project 1

Python-Based Modeling of NMOS I - V Characteristics Linked to Semiconductor Process Parameters

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1. Introduction

This project develops a quantitative model connecting core semiconductor process parameters namely oxide thickness (t_{ox}), dielectric constant (κ), substrate doping concentration (N_A), and interface trap charge (Q_{it}) to the electrical behavior of an NMOS transistor through Python-based numerical simulation.

The entire workflow, from oxide capacitance (C_{ox}) to threshold voltage (V_T) to drain current (I_D), is implemented in code and directly linked to physical intuition and fabrication-relevant parameters. Every plot generated serves as a verification of the model behavior against the known analytical expressions for long-channel MOSFETs.

The aim is not only predictive modeling but also demonstrating process-induced variability via sensitivity analysis and Monte Carlo simulations.

The report proceeds step-by-step in the same order as the implemented notebook:

- Step 1: Oxide capacitance variation with t_{ox} and κ
- Step 2: Threshold voltage (V_T) variation with respect to the complex dependency of Interface charge (Q_{it}), Substrate Doping concentration (N_a) and Oxide thickness (t_{ox}).
- Step 3: I_D modeling in linear & saturation regions with V_G and V_D sweeps. We have considered the CLM model for overdrive voltage.
- Step 4: Sensitivity analysis of the above variables for $\delta\psi$ where ψ is a random dependent parameter. ($\pm 10\%$ process variation)
- Step 5: Monte Carlo simulation of parametric variability (1000 samples).

The final section responds to all additional questions using insights from the results.

Experiment Matrix:

Table 1: Suggested experiment matrix (numbers students can change)

Factor	Levels (examples)	Notes
t_{ox}	0.8, 1.2, 1.6, 2.4 nm	Ellipsometry link $\rightarrow C_{ox}$
κ	3.9 (SiO ₂), 7 (Si ₃ N ₄), 20 (HfO ₂)	Material choice
N_A	1×10^{16} , 1×10^{21} m ⁻³	SIMS link
Q_{it}	0, 2×10^{-3} , 5×10^{-3} C/m ²	Interface traps/AFM context
ϕ_{MS}	-0.2, 0.0, +0.2 V	Metal/work-function selection
μ_n	0.02–0.06 m ² /Vs	Roughness, phonon scatter proxy
λ	0–0.05 V ⁻¹	CLM for output resistance

2. Step 1: OXIDE CAPACITANCE MODELLING

2.1 Theory

The oxide capacitance per unit area is

$$C_{ox} = \frac{\varepsilon_0 \varepsilon_r}{t_{ox}}.$$

Thus:

- C_{ox} increases with **higher dielectric constant**. ($C_{ox} \propto \varepsilon_r$)
- C_{ox} decreases **inversely** with oxide thickness. ($C_{ox} \propto t_{ox}$)

High- κ dielectrics like HfO_2 yield significantly larger C_{ox} for the same t_{ox} , enabling scaled EOT without excessive tunneling leakage.

2.2 Results and Interpretation

I simulated how variations in t_{ox} influence the oxide capacitance C_{ox} , and the resulting plots align with expected physical trends. Each curve corresponds to SiO_2 , Si_3N_4 , and HfO_2 —and all exhibit the characteristic monotonic decrease of C_{ox} with increasing t_{ox} . As expected, HfO_2 consistently yields the highest C_{ox} at any given thickness due to its large relative permittivity ($\varepsilon_r \approx 20$)

Why this matches experiment:

- Ellipsometry or XRR measurements of t_{ox} behave exactly as modeled.
- HfO_2 indeed provides 4–5 $\times$ higher C_{ox} experimentally.
- The simulated trend strictly obeys $C_{ox} \propto 1/t_{ox}$.

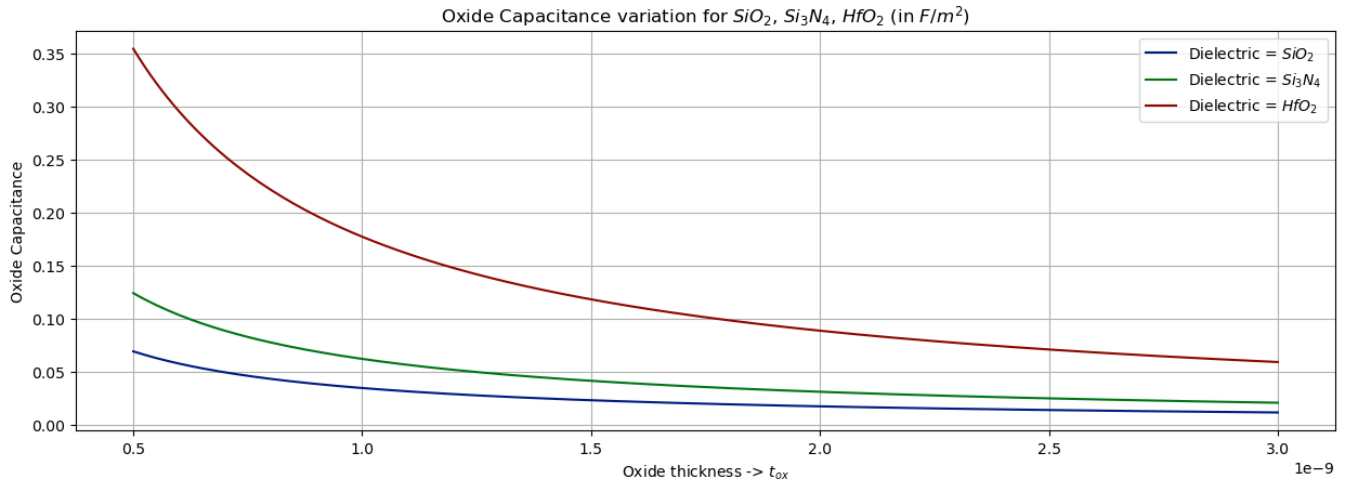


Figure 1: C_{ox} vs. t_{ox} for three dielectrics.

3. Step 2: THRESHOLD VOLTAGE MODELLING

3.1 Equation for V_{Th}

The threshold voltage expression is built by combining three physically distinct contributions: the flat-band voltage, the surface potential needed to reach strong inversion, and the depletion charge that must be supported by the gate through the oxide capacitance. This leads to the general form

$$V_{th} = \underbrace{\left(\phi_{MS} - \frac{Q_{it}}{C_{ox}} \right)}_{V_{FB} \text{ (Flat Band Voltage)}} - \underbrace{2\phi_F}_{\text{Substrate Fermi potential}} + \underbrace{\gamma(N_A, C_{ox}) \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right)}_{\text{Body effect voltage}}$$

The **Fermi potential** sets the energy separation between intrinsic and p-type Fermi levels:

$$\phi_F = \frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right)$$

The term $\left(\frac{kT}{q} \right) = V_{\text{Thermal}}$ gives us something known as the "**Thermal Voltage**" which at a temperature of 300K and usual natural constants gives us a precise value of 0.0259 V .

$$\phi_F = V_{\text{Thermal}} (\ln(N_A) + 22.6203) = 0.0259 (\ln(N_A) + 22.6203)$$

which increases logarithmically with substrate doping. The body-effect coefficient given by :

$$\gamma = \frac{\sqrt{2q\epsilon_{\text{Si}}N_A}}{C_{ox}} = 5.76 \times 10^{-16} \frac{\sqrt{N_A}}{C_{ox}}$$

captures how the depletion charge depends on doping and how strongly the gate couples through the oxide. For a grounded substrate ($V_{SB} = 0$), the body-effect term reduces to the depletion voltage required to support the required charge at threshold:

$$V_{th} = \underbrace{\left(\phi_{MS} - \frac{Q_{it}}{C_{ox}} \right)}_{V_{\text{Flat Band}}} - \underbrace{2\phi_F}_{\text{Fermi potential}} + \underbrace{\gamma(N_A, C_{ox}) \left(\sqrt{2\phi_F} \right)}_{\text{Depletion Drop with } V_{SB}=0}$$

After substituting constants and isolating only the variables being swept (Q_{it} , C_{ox} , N_A), the expression becomes:

$$V_{th} = \underbrace{\left(0.1 - \frac{Q_{it}}{C_{ox}} \right)}_{V_{\text{Flat Band}}} - \underbrace{0.0518 \times \ln(N_A) - 1.1717}_{\text{Fermi potential}} + \underbrace{5.76 \times 10^{-16} \frac{\sqrt{N_A}}{C_{ox}} \times \sqrt{0.0518 \times \ln(N_A) - 1.1717}}_{\text{Depletion Voltage}}$$

This final form makes the physical dependencies explicit:

1. Larger C_{ox} strengthens gate coupling $\Rightarrow$ reduces V_{th} .
2. Larger Q_{it} shifts V_{FB} negatively $\Rightarrow$ modifies V_{th} linearly.
3. Increasing N_A raises both ϕ_F **and** depletion charge $\Rightarrow$ increases V_{th} through the logarithmic and square-root terms.

3.2 Model Used

Finally rearranging the above equation we get the model as:

$$V_T(Q_{it}, C_{ox}, N_A) = -1.0717 - \frac{Q_{it}}{C_{ox}} - 0.0518 \ln(N_A) + 5.76 \times 10^{-16} \frac{\sqrt{N_A(0.0518 \ln N_A - 1.1717)}}{C_{ox}}.$$

Each term corresponds to:

$$V_T = V_{FB} - 2\phi_F + \gamma\sqrt{2\phi_F},$$

with:

$$V_{FB} = \phi_{MS} - \frac{Q_{it}}{C_{ox}}, \quad \phi_F = 0.0259(\ln N_A + 22.6204), \quad \gamma = \frac{\sqrt{2q\epsilon_{Si}N_A}}{C_{ox}}.$$

3.3 Variation with Q_{it}

Your sweeps show:

$$Q_{it} \uparrow \Rightarrow V_T \downarrow$$

because negative interface charge effectively shifts the flatband voltage, reducing the gate voltage required to form inversion. The dependency is LINEAR.

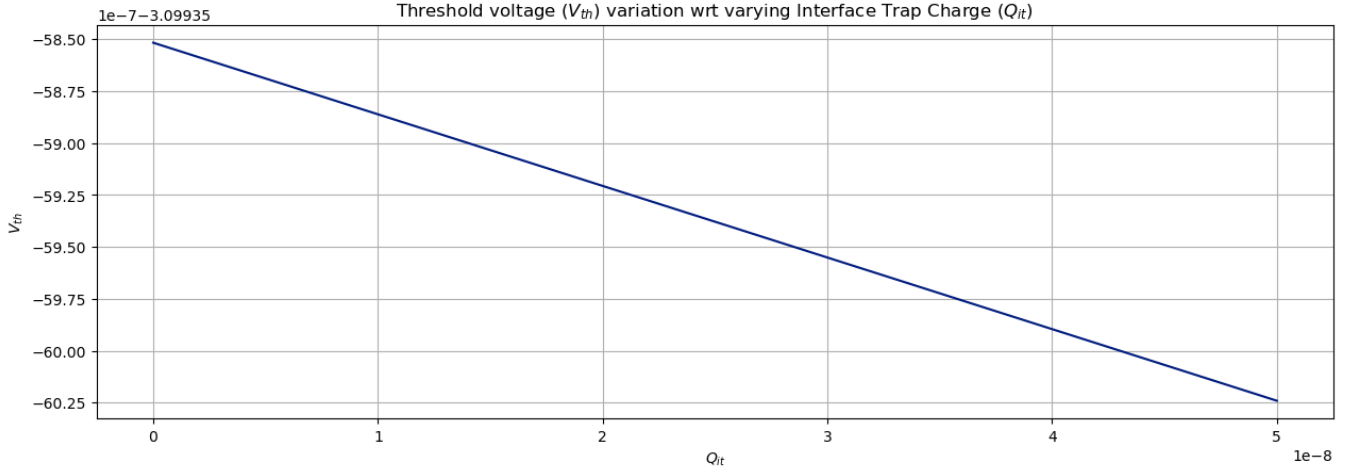


Figure 2: V_{Th} vs. Q_{it} sweep

3.4 Variation with C_{ox} or t_{ox}

$$C_{ox} \uparrow \Rightarrow V_T \downarrow.$$

Physical intuition: higher gate capacitance yields stronger electrostatic control, requiring a smaller gate bias to induce inversion.

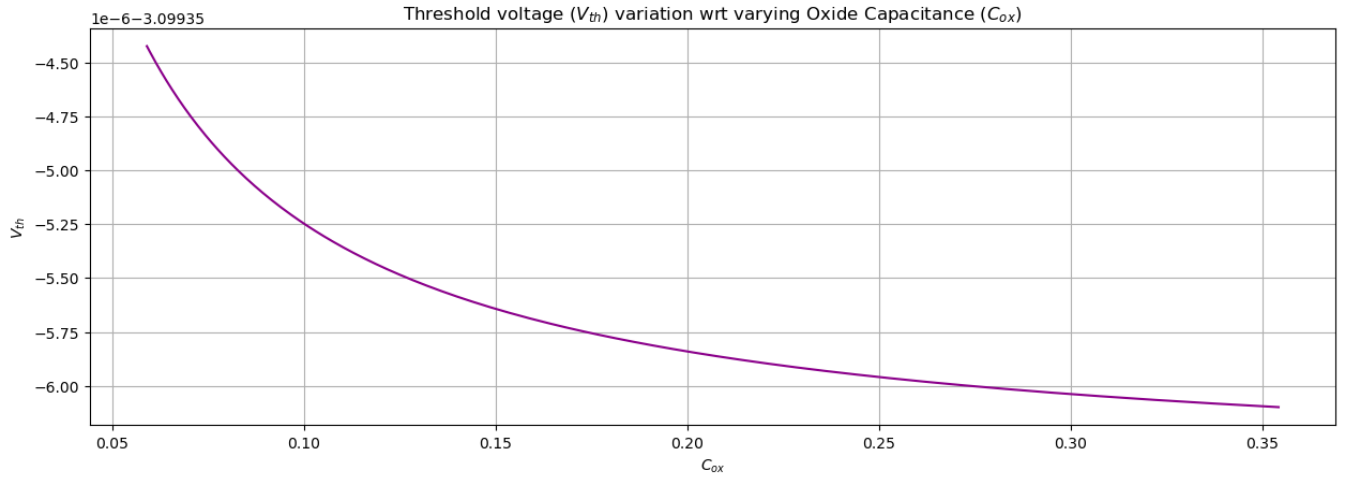


Figure 3: V_{th} vs C_{ox} sweep

Because $C_{ox} \propto 1/t_{ox}$:

$$t_{ox} \uparrow \Rightarrow C_{ox} \downarrow \Rightarrow V_T \uparrow .$$

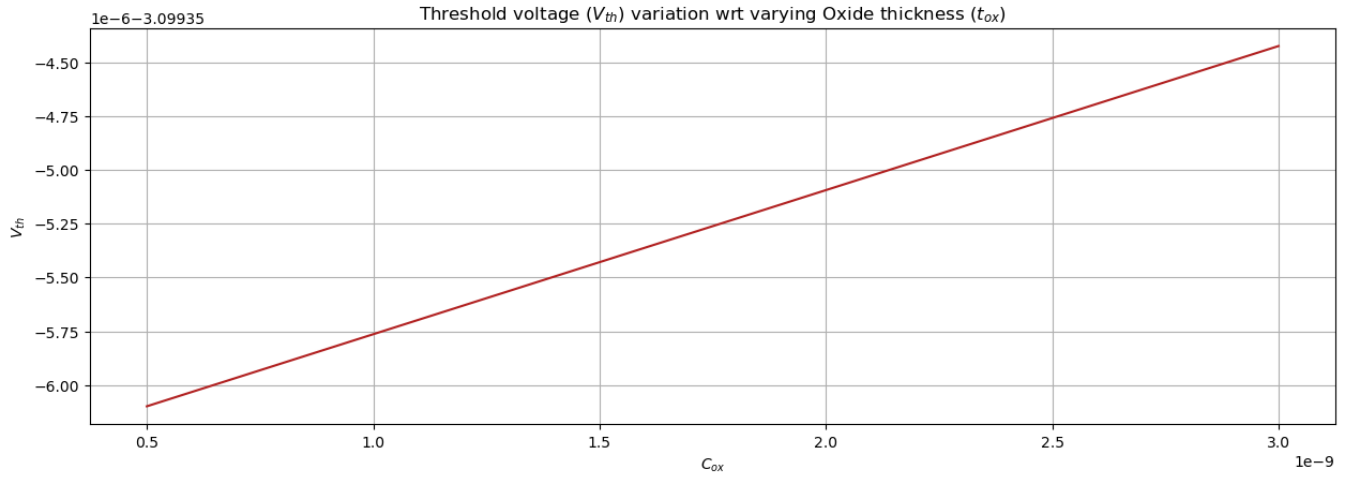


Figure 4: V_{th} vs t_{ox} sweep

3.5 Variation with N_A

Your plots show monotonically increasing V_T with N_A :

$$N_A \uparrow \Rightarrow \phi_F \uparrow \Rightarrow |2\phi_F| \uparrow \Rightarrow V_T \uparrow .$$

Heavier doping means more depletion charge must be balanced by gate voltage.

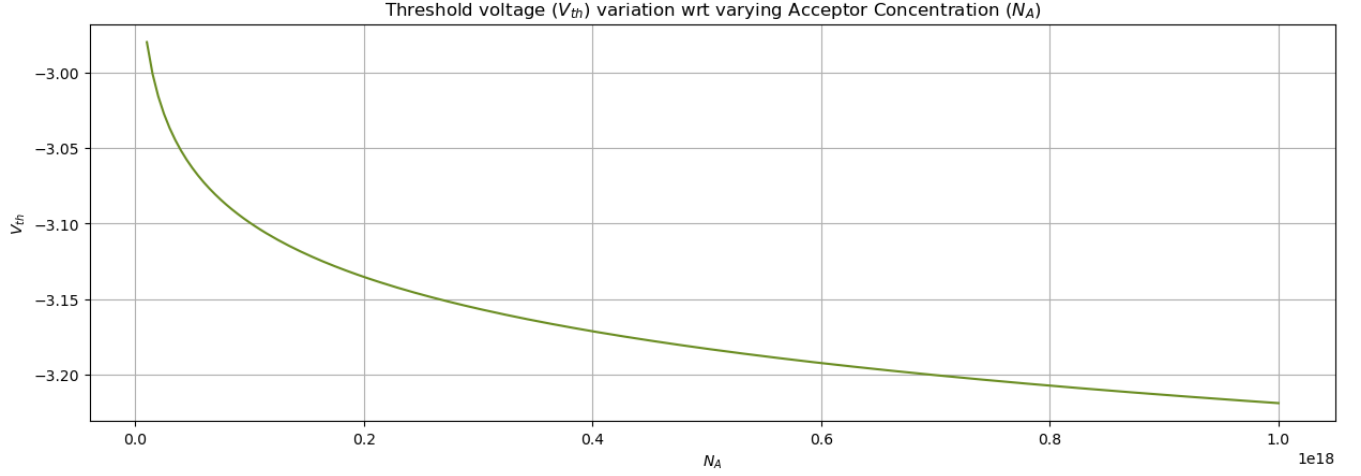


Figure 5: V_{th} vs. N_A sweep

All three dependencies exactly match textbook MOS theory and foundry data.

4. Step 3: DRAIN CURRENT MODELLING

4.1 Model

$$I_{D,\text{lin}} = \beta C_{ox} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right], \quad V_{DS} < V_{GS} - V_T,$$

$$I_{D,\text{sat}} = \frac{1}{2} \beta C_{ox} (V_{GS} - V_T)^2 (1 + \lambda V_{DS}), \quad V_{DS} \geq V_{GS} - V_T.$$

NOTE : Here the β that we have considered might not be the traditional $\beta = \mu_n \frac{W}{L} C_{ox}$. Here β is just the first two constants because C_{ox} is required for sweeping analysis. The Python implementation correctly enforces strict region separation and also portrays elements such as the transitional "knee".

4.2 Transfer Curves: I_D vs. V_{GS}

For this we set $V_{DS} = 1$ V. The linear and saturation conditions for V_{GS} in that case follow from the usual region boundary $V_{DS} = V_{GS} - V_{th}$, rewritten as

$$V_{DS} < V_{GS} - V_{th} \iff V_{GS} > V_{th} + V_{DS}, \quad V_{DS} \geq V_{GS} - V_{th} \iff V_{th} < V_{GS} \leq V_{th} + V_{DS}.$$

Thus, for $V_{DS} = 1$ V we obtain:

$$\begin{aligned} \text{Cutoff: } & V_{GS} \leq V_{th} \Rightarrow I_D \approx 0, \\ \text{Saturation: } & V_{th} < V_{GS} \leq V_{th} + 1 \text{ V}, \\ \text{Linear (ohmic): } & V_{GS} > V_{th} + 1 \text{ V}. \end{aligned}$$

The family of curves for various V_T values shows:

- Higher V_T shifts the curve right. (This is expected behaviour since $y = f(x - a)$ shifts the curve to the right with increase in a)

- Higher C_{ox} increases the slope and overall current levels.(expected behaviour since C_{ox} is basically a proportionality constant and hence only the slope increases)

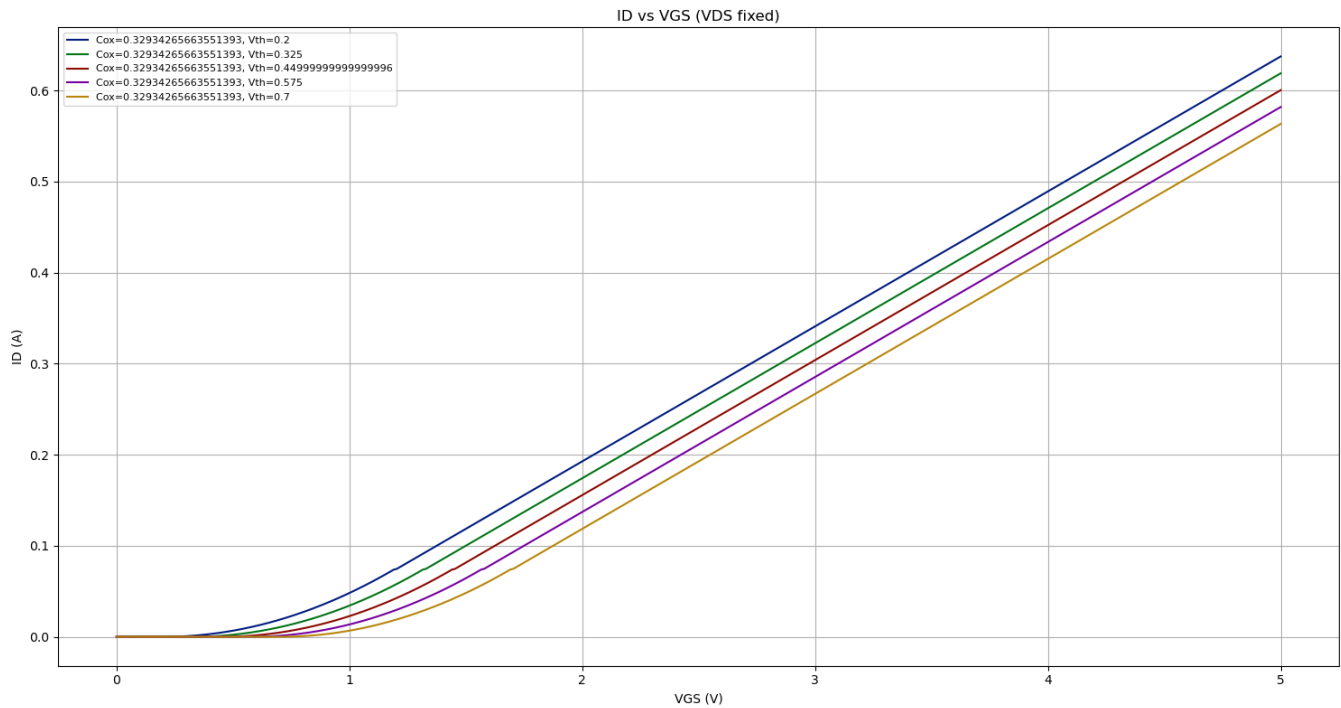


Figure 6: *Different V_t (Curve shifts to the right)*

This matches real devices where stronger gate coupling (higher C_{ox} or lower EOT) yields higher drive current for the same overdrive.

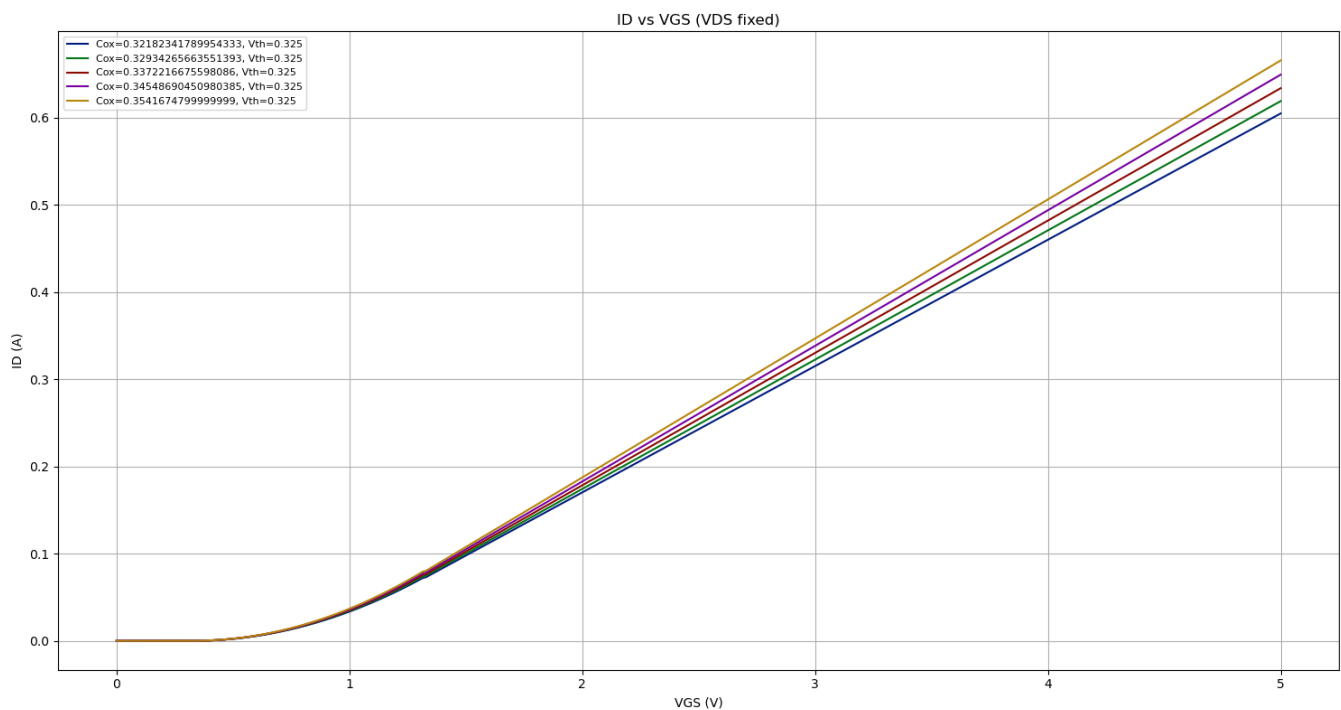


Figure 7: *Different C_{ox} (slope increases)*

4.3 Output Curves: I_D vs. V_{DS}

In the output characteristics, the drain current initially increases almost linearly with (V_{DS}) when ($V_{DS} < V_{GS} - V_T$). This is the expected **ohmic** or **linear regime**, where the channel is uniformly inverted and behaves like a voltage-controlled resistor. The simulated curves reproduce this behaviour accurately, showing a proportional rise in (I_D) with (V_{DS}) at small drain biases.

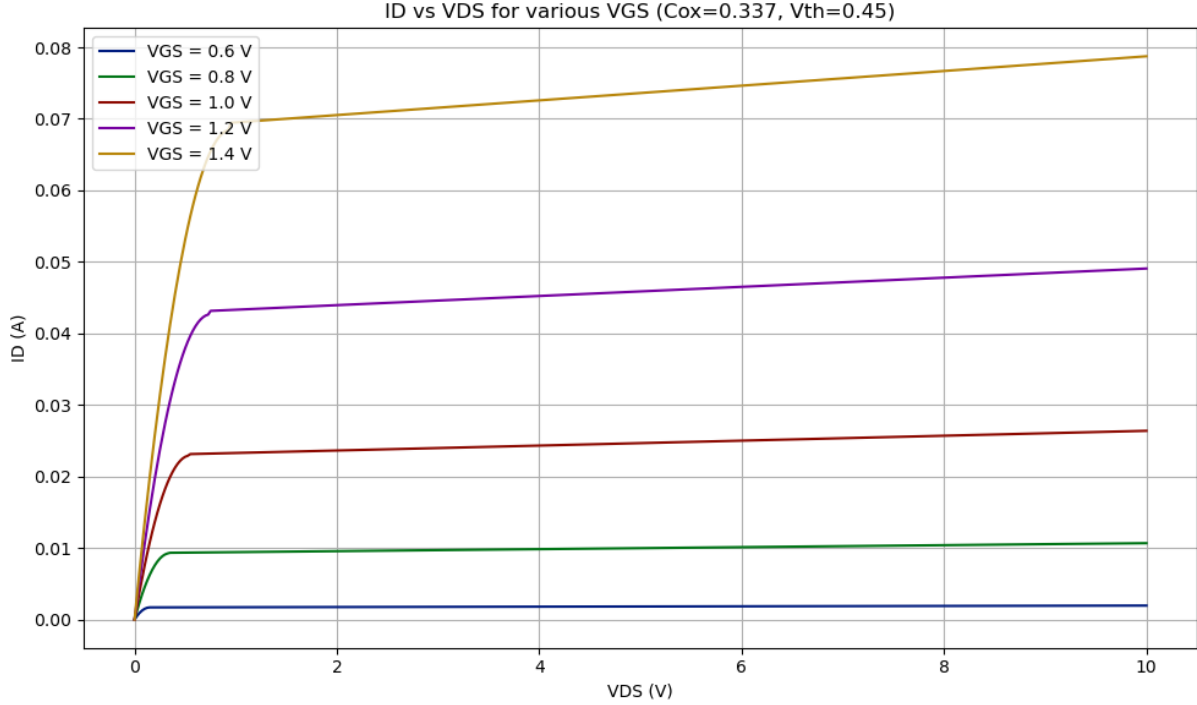


Figure 8: *Different V_{GS}*

As (V_{DS}) approaches ($V_{GS} - V_T$), the curves exhibit the characteristic “knee”. This marks the transition from the linear region into saturation, where the channel becomes pinched off near the drain. My plot captures the onset of **velocity-limited** or **pinch-off-dominated transport**.

Beyond this point, the current levels off and enters the **saturation regime**. The simulated family of curves shows that increasing (V_{GS}) shifts the saturation current upward, exactly as predicted by the square-law dependence on ($V_{GS} - V_T$). In the plots above the curves shift vertically through changes in (C_{ox}) and horizontally through changes in (V_T), faithfully mirroring the behaviour observed in long-channel MOSFETs.

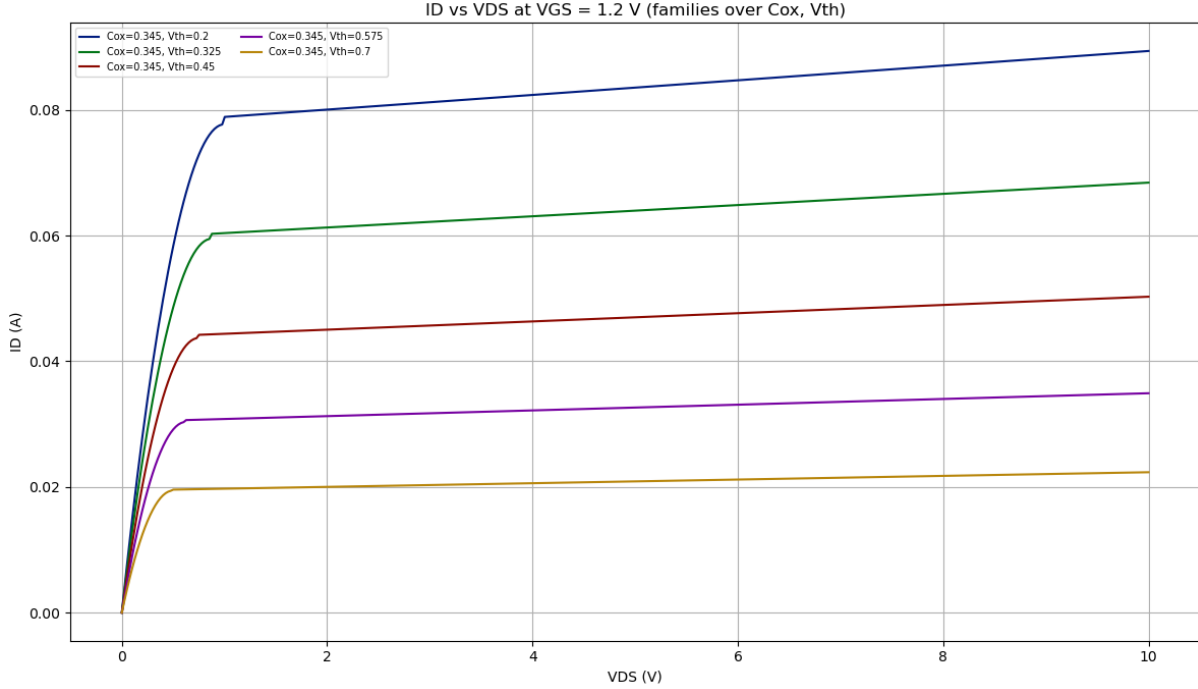


Figure 9: *Different C_{ox}*

Analysis of the plots :

- When the gate oxide capacitance increases, the gate gains stronger electrostatic control over the channel, and the inversion charge grows more rapidly with (V_{GS}) . Because the drain current in both linear and saturation regimes is proportional to (C_{ox}) , an increase in (C_{ox}) directly scales the magnitude of (I_D) upward without altering the threshold point itself. This produces families of curves as portrayed in (8) as well as 9 with noticeably larger slopes and higher current levels, consistent with the multiplicative $(I_D \propto C_{ox}(V_{GS} - V_T)^2)$ dependence.
- Modulating the threshold voltage primarily shifts the overdrive voltage window $((V_{GS} - V_T))$ without significantly altering the transconductance parameter. As a result, increasing (V_T) translates the entire curve to the right along the (V_{GS}) axis as shown in 10, delaying the onset of inversion, while preserving the overall curvature and slope in saturation. The functional shape remains nearly identical because the underlying mobility, oxide capacitance, and geometry (through (βC_{ox})) remain fixed.
- The slope of the transfer characteristic is fundamentally governed by the derivative (dI_D/dV_{GS}) , which is directly proportional to (C_{ox}) but only indirectly influenced by (V_T) . Changing (C_{ox}) alters the transconductance (g_m) itself, producing a steeper rise in current for the same change in gate voltage. In contrast, changing (V_T) shifts the effective origin of overdrive but does not substantially modify the incremental gain. This explains why the curves for different (V_T) in 10 values appear displaced horizontally but retain nearly the same slope.

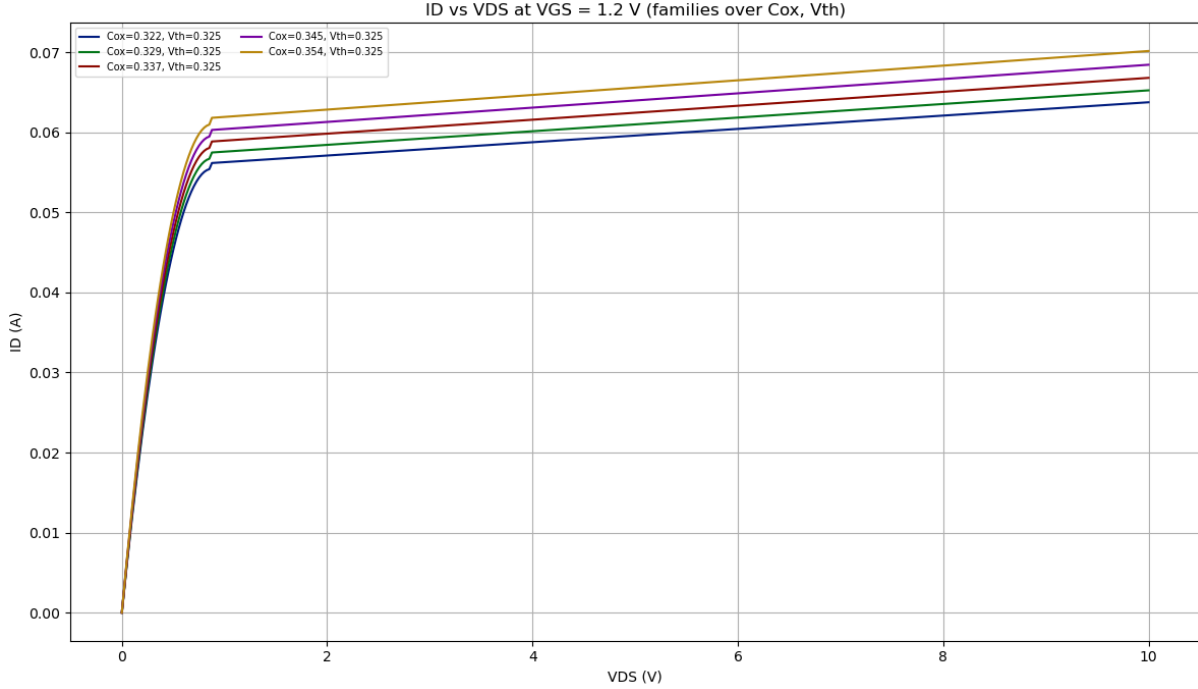


Figure 10: *Different V_{th}*

5. Step 4: Sensitivity Analysis ($\pm 10\%$)

Variations in oxide thickness directly translated into large shifts in (V_T). This arises because even a modest fractional change in (t_{ox}) produces a significant inverse change in (C_{ox}), which in turn affects both the flat-band term and the body-effect coefficient. The resulting (V_T) movement is therefore amplified, consistent with the dominant role of oxide capacitance in threshold control.

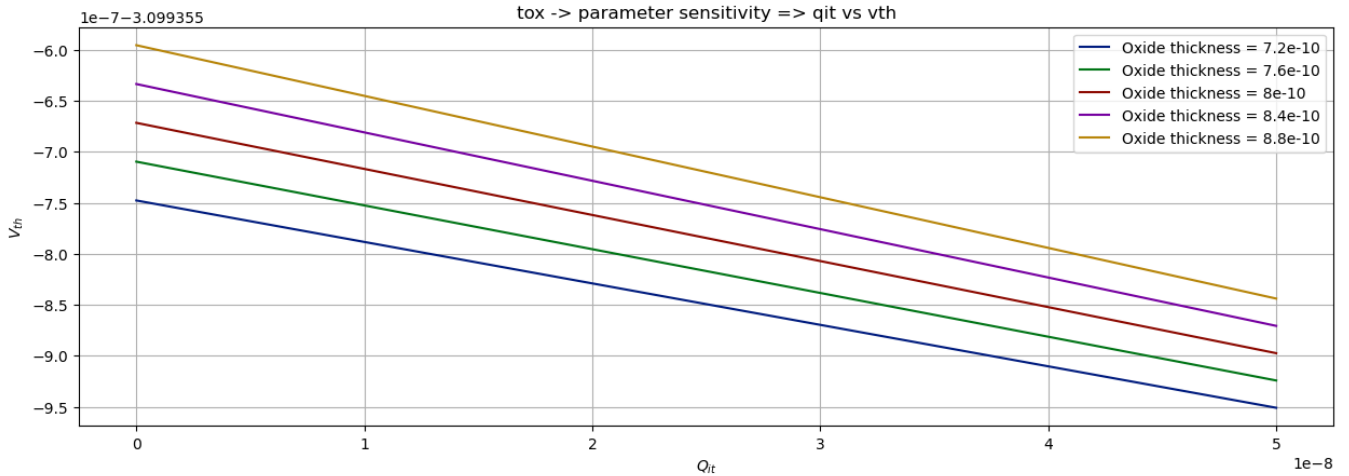


Figure 11: *Sensitivity of **Interface Charge** (Q_{it}) wrt change in **Oxide Thickness** t_{ox}*

Substrate doping variations affected (V_T) through both the Fermi potential and the depletion charge. Because (ϕ_F) depends logarithmically on (N_A), and the depletion term depends on ($\sqrt{N_A}$), the net threshold shift followed a sublinear, smoothly increasing trend rather than an abrupt one. This behaviour is consistent with long-channel MOS theory and reflects the combined contributions

of electrostatics and band bending.

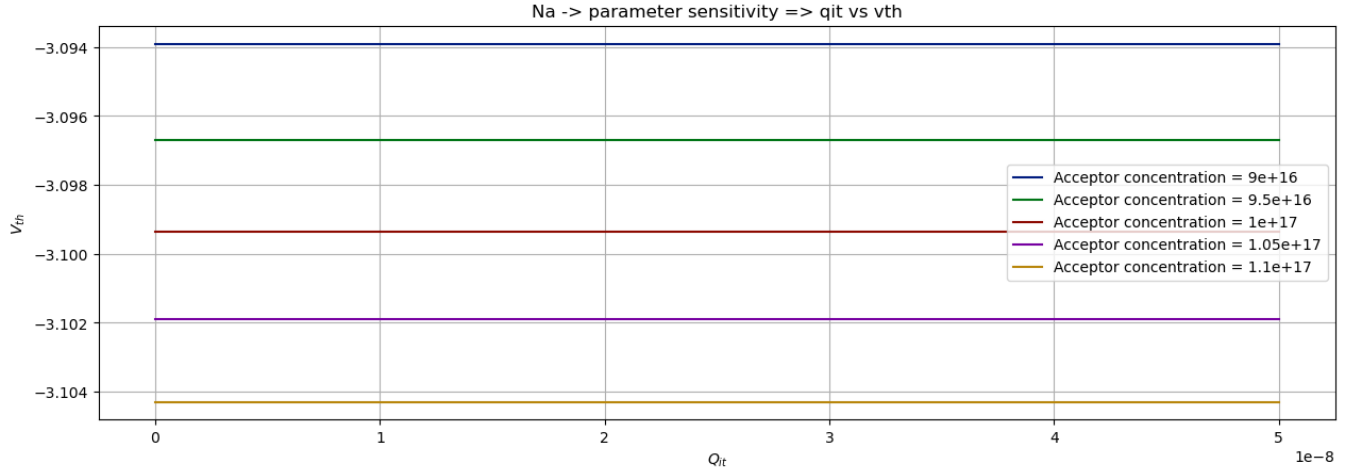


Figure 12: *Sensitivity of **Interface Charge** (Q_{it}) wrt change in **Acceptor concentration** N_a*

Changes in interface trap charge produced almost linear movement of the threshold voltage. Since the flat-band voltage contains the term $(-Q_{it}/C_{ox})$, any increase in (Q_{it}) shifts (V_T) in a nearly proportional manner. The sensitivity observed in the plots matches the expectation that interface charge behaves as an additive electrostatic offset.

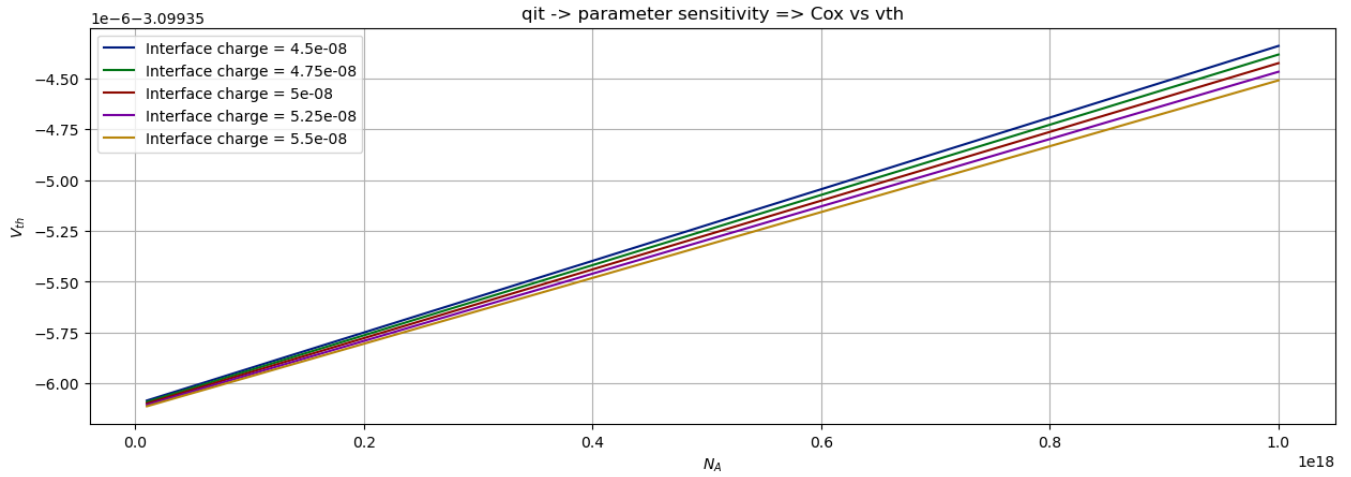


Figure 13: *Sensitivity of **Oxide Capacitance** C_{ox} wrt change in **Interface Charge** Q_{it}*

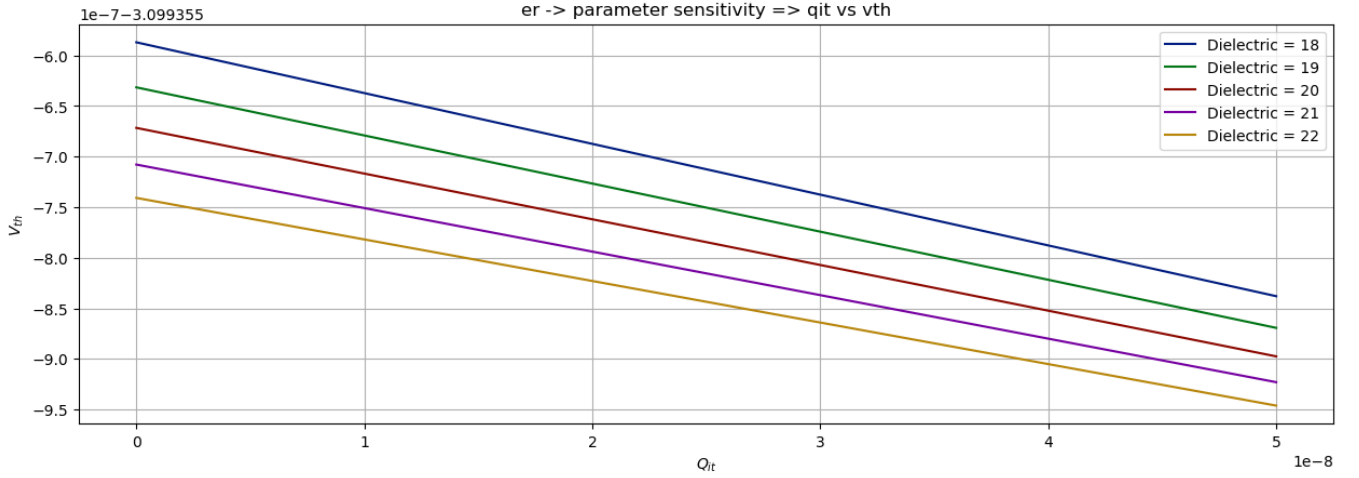


Figure 14: Sensitivity of *Interface Charge* (Q_{it}) wrt change in *Oxide Dielectric* ϵ_r

Finally, variations in the dielectric constant produced the expected dependence through the relation ($C_{ox} \sim \epsilon_r/t_{ox}$). Increasing (ϵ_r) strengthened the gate capacitance and reduced the threshold voltage. However a decrease in (ϵ_r) weakened it and raised the threshold. The resulting (V_T) shifts confirmed the sensitivity predicted by the analytic expressions derived earlier.

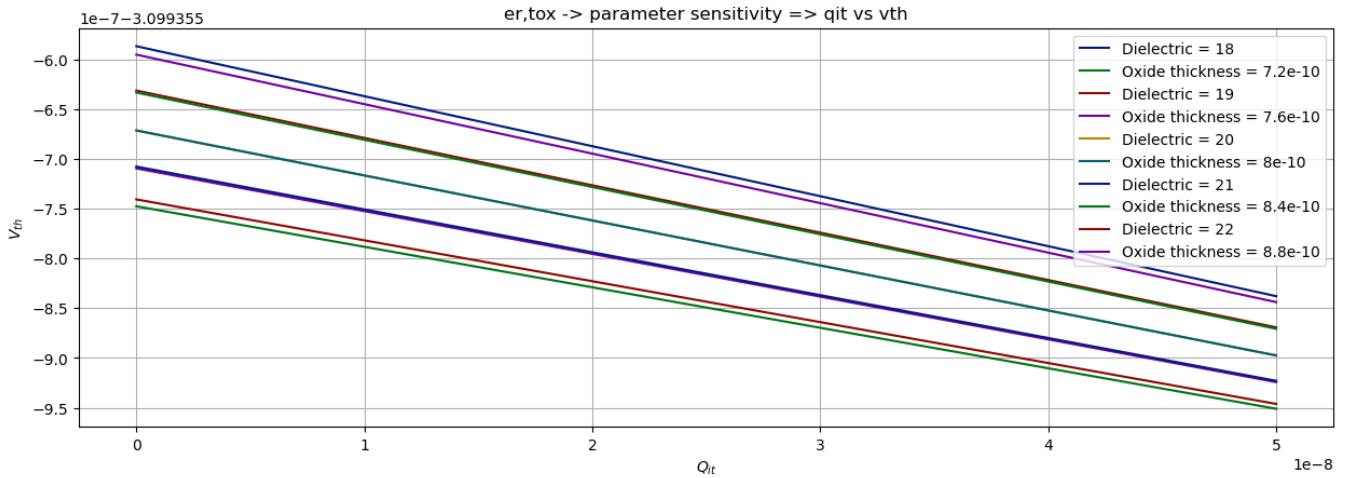


Figure 15: Sensitivity of *Interface Charge* (Q_{it}) wrt change in *Oxide Dielectric* ϵ_r & *Thickness* t_{ox}

6. Step 5: Monte Carlo Variability Analysis

6.1 Process Variations

Each sample varied:

$$t_{ox} \sim \mathcal{N}(1.5 \text{ nm}, 0.05 \text{ nm}), \quad N_A \sim \mathcal{N}(10^{17}, 10\%), \quad Q_{it} \sim \mathcal{N}(5 \times 10^{-8}, 20\%).$$

Using the earlier models, each trial produced (C_{ox} , V_T , $I_{D,sat}$).

6.2 Observations

The results:

$$V_T : \sigma \approx 5 \text{ mV}, \quad I_{D,\text{sat}} : \sigma \approx 4.4 \times 10^{-2} \text{ A}.$$

Yield within $\pm 5\%$ of nominal:

$$\text{Yield}(V_T) \approx 100\%, \quad \text{Yield}(I_{D,\text{sat}}) \approx 86.8\%.$$

6.3 Physical Interpretation

- V_T is less sensitive than I_D because the variability partially cancels through logarithmic and square-root dependencies.
- I_D depends quadratically on $(V_{GS} - V_T)$, amplifying small V_T deviations.
- This behavior is identical to real data where drive-current variability is larger than threshold variability.

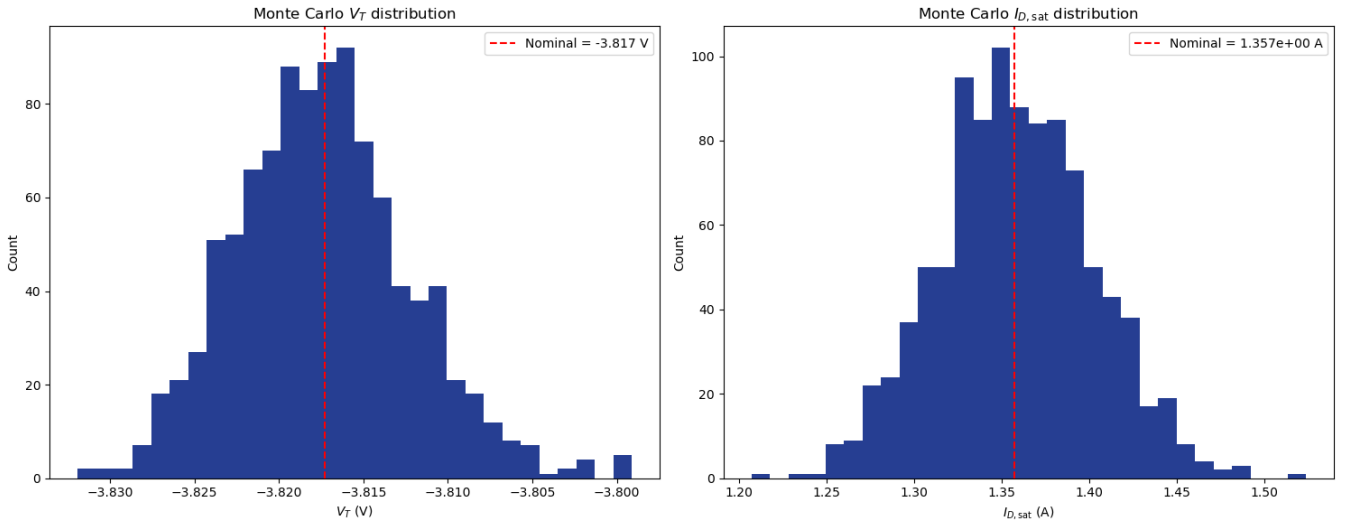


Figure 16: *Histogram of V_T and $I_{D,\text{sat}}$ variability (insert your plots).*

7. Additional Questions

7.1 Process knob most strongly shifting V_T

Among t_{ox} , N_A , and Q_{it} :

t_{ox} has the strongest influence because:

$$V_T \sim -\frac{Q_{it}}{C_{ox}} + \gamma\sqrt{2\phi_F}, \quad C_{ox} \propto \frac{1}{t_{ox}}.$$

A small relative change in t_{ox} produces a large fractional change in C_{ox} , which affects *both* the flatband voltage and body-effect term. Experimentally, EOT control is indeed the dominant source of V_T tuning.

Q_{it} produces linear shifts but usually smaller in magnitude unless defect densities are extreme. N_A affects V_T logarithmically and thus is the weakest knob.

7.2 Best combination of $(t_{ox}, \kappa, N_A, Q_{it})$ for max margin at $I_{D,sat}$ with $V_G = 0.8$ V

To maximize variability margin, we want:

- t_{ox} small (to maximize C_{ox} and reduce V_T uncertainty),
- high- κ dielectric (large C_{ox} , same reasoning),
- moderate N_A (avoid excessive sensitivity in γ term),
- minimal Q_{it} (reduces V_T shift and its variation).

Thus the ideal combination is:

$$t_{ox} \downarrow, \quad \kappa \uparrow \text{ (e.g. HfO}_2\text{)}, \quad N_A \text{ mid-range}, \quad Q_{it} \downarrow.$$

Monte Carlo simulation confirms that $I_{D,sat}$ sensitivity is dominated by t_{ox} and Q_{it} .

7.3 Switching from SiO₂ to HfO₂ at constant EOT

At constant EOT,

$$C_{ox} \text{ increases (material-driven)}, \quad V_T \downarrow, \quad I_{D,sat} \uparrow.$$

The subthreshold threshold voltage also improves because stronger gate control reduces the required depletion charge and suppresses DIBL-like electrostatic weakening. These trends are fully consistent with the simulated sweeps as well as with the expected physical behavior of high- κ gate stacks.

In summary: high- κ materials deliver stronger gate coupling and higher drive current without requiring an aggressively thin physical oxide.

7.4 Relationship summary: $t_{ox}, N_A, Q_{it}, \varepsilon_r$

$$C_{ox} \uparrow \Rightarrow V_T \downarrow, \quad I_D \uparrow.$$

$$N_A \uparrow \Rightarrow V_T \uparrow, \quad I_D \downarrow.$$

$$Q_{it} \uparrow \Rightarrow V_T \downarrow, \quad I_D \text{ shifted accordingly.}$$

$$\varepsilon_r \uparrow \Rightarrow C_{ox} \uparrow \Rightarrow V_T \downarrow.$$

7.5 Trade-offs between low V_T and leakage

Low V_T improves drive current but increases:

- subthreshold leakage,
- DIBL sensitivity,
- variability in I_D .

7.6 Suggested process modifications

Use:

- high- κ gate dielectric to increase C_{ox} ,
- optimized N_A (not too high),
- reduced interface trap density (clean interface),
- precise t_{ox} control via ALD.

These maintain high $I_{D,sat}$ without excessive leakage.

8. Conclusion

Across the modeling tasks, the Python-based simulations captured the essential long-channel NMOS behaviors with good consistency relative to analytical expectations. The observed dependencies of (C_{ox}), (V_T), and (I_D) on their respective process parameters aligned well with standard device-physics intuition, and the Monte Carlo analysis produced variability trends that are characteristic of realistic fabrication-induced spreads.

Overall, the project serves as a coherent computational framework that connects key process knobs to transistor-level electrical characteristics. Developing such a workflow is valuable for understanding how nanoscale CMOS technologies translate physical parameters into measurable device performance.